



# Dielectric Properties

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## **Definition of dielectric behavior:**

Transmitting an electric force without conduction

## **Definition of dielectric material:**

A material with dielectric properties:

An insulator but able to transmit an electric field

Some dielectric materials can be polarized

# Capacitance

- Capacitance,  $C$ , is the ability to store electric charge

$$C = \frac{Q}{V} \quad \text{units of coulombs / V or Farads (F)}$$

$Q$  = quantity of charge

$V$  = voltage

- for a parallel plate capacitor

$$C = \epsilon_0 \frac{A}{l}$$

$A$  = area

$l$  = distance over voltage applied

$\epsilon_0$  = the permittivity of vacuum (or permittivity of free space)

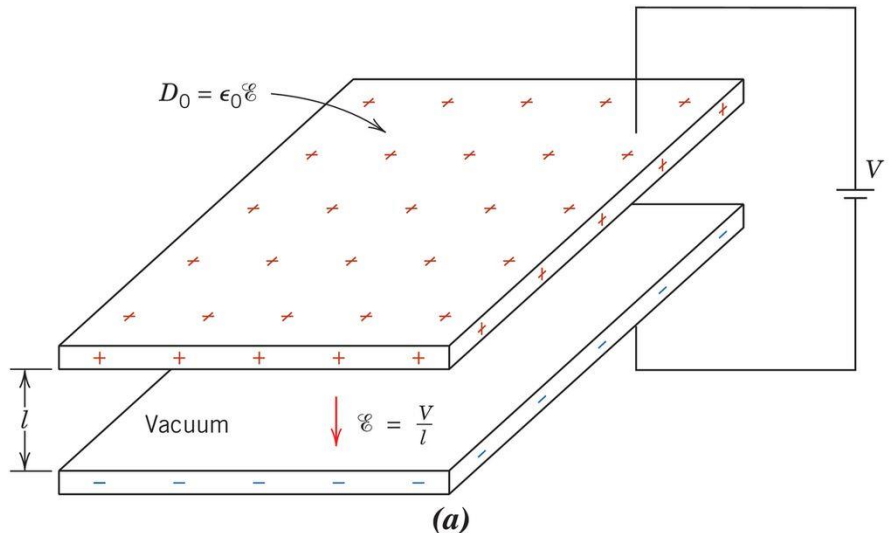


Fig 19.28a Callister & Rethwisch 9e

# Permittivity

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- **Permittivity** is defined as the ability of a material to store electrical energy in an electric field

$\epsilon_0$ , the **permittivity of vacuum** (or permittivity of free space) is a universal constant =  $8.85 \times 10^{-12}$  F/m

- **Relative permittivity** is defined for a material as the ratio of the permittivity of the material to the permittivity of vacuum

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} \quad \text{Also called **dielectric constant**, } k. \quad \text{No units}$$

- Insulating materials have  $\epsilon_r$  that is  $> 1$   
Those that can polarize have  $\epsilon_r$  that is  $\gg 1$

# Dielectric strength

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- **Dielectric strength** the strength of an electric field required to cause breakdown

If the electric field strength is large enough to excite enough electrons across the band gap, **electrical breakdown** can occur. The larger band gap, the greater the dielectric strength.

The **units** of dielectric strength are V/m  
with values typically of MV/m

Some American texts (including Callister) use units of V/mil  
1 mil = 0.025mm (1/1000 inches)

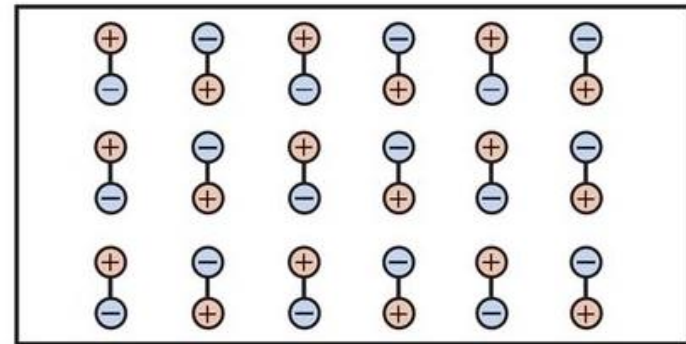
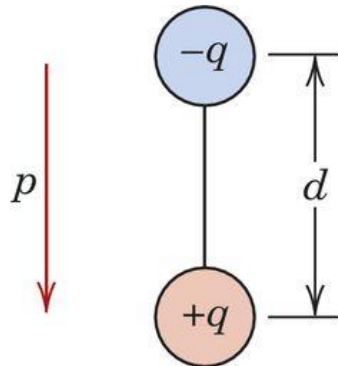
# Dielectric properties

| Material                         | Dielectric Constant |           | Dielectric Strength (V/mil) <sup>a</sup> |
|----------------------------------|---------------------|-----------|------------------------------------------|
|                                  | 60 Hz               | 1 MHz     |                                          |
| Ceramics                         |                     |           |                                          |
| Titanate ceramics                | —                   | 15–10,000 | 50–300                                   |
| Mica                             | —                   | 5.4–8.7   | 1000–2000                                |
| Steatite (MgO–SiO <sub>2</sub> ) | —                   | 5.5–7.5   | 200–350                                  |
| Soda–lime glass                  | 6.9                 | 6.9       | 250                                      |
| Porcelain                        | 6.0                 | 6.0       | 40–400                                   |
| Fused silica                     | 4.0                 | 3.8       | 250                                      |
| Polymers                         |                     |           |                                          |
| Phenol-formaldehyde              | 5.3                 | 4.8       | 300–400                                  |
| Nylon 6,6                        | 4.0                 | 3.6       | 400                                      |
| Polystyrene                      | 2.6                 | 2.6       | 500–700                                  |
| Polyethylene                     | 2.3                 | 2.3       | 450–500                                  |
| Polytetrafluoroethylene          | 2.1                 | 2.1       | 400–500                                  |
| Air                              | 1                   | 1         | 75                                       |

# Polarization

- capacitance is greatly increased by using dielectric materials that can polarize

Where charges are separated in a material a dipole moment is created, characterized by the polarization vector  $\mathbf{p} = q\mathbf{d}$

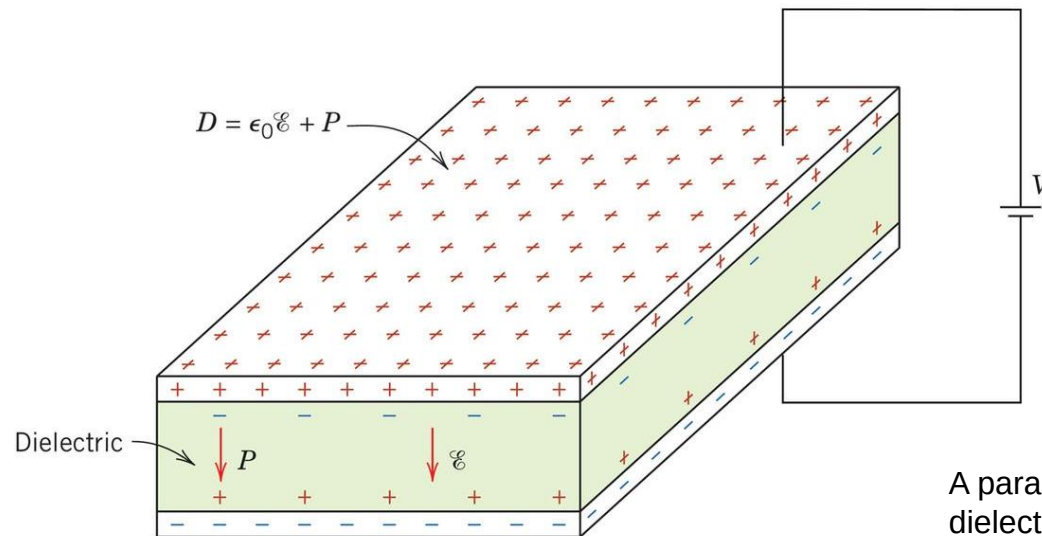


Within the solid the dipoles will be arranged so that there is no net charge on the material

Fig 19.29 and 19.31 Callister & Rethwisch 9e

# Polarization

In an electric field the dipoles will align with the electric field vector (as much as is possible). This creates a net charge to the surface.



A parallel-plate capacitor when a dielectric material is present.  
Fig 19.28b Callister & Rethwisch 9e

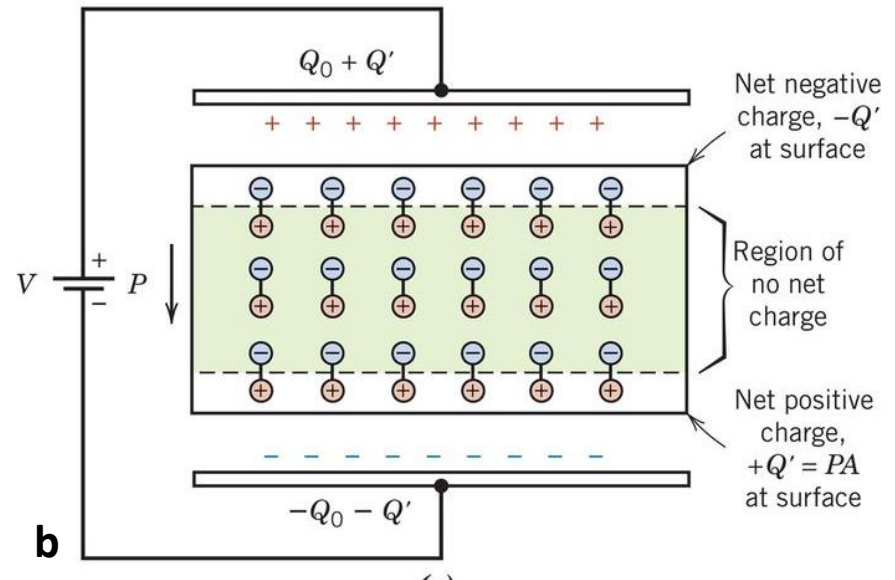
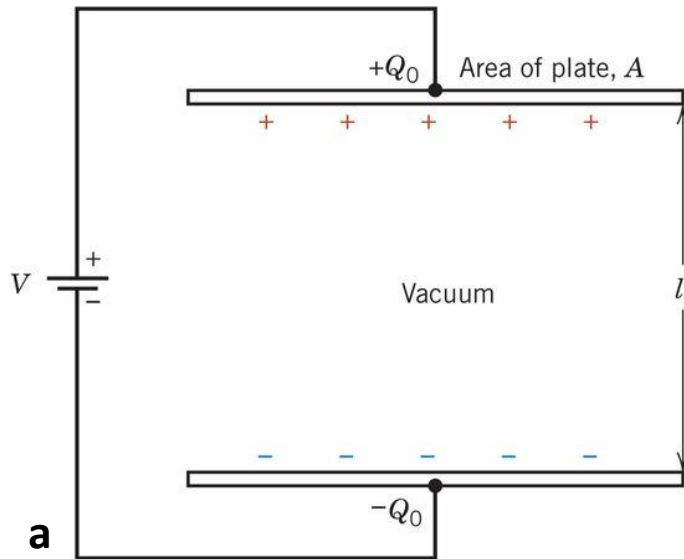
When a dielectric material is placed between the plates (instead of air), more charge can be stored in the capacitor



# Dielectric displacement

- **dielectric displacement,  $D$** , is a measure of the surface charge density in a capacitor (quantity of charge / area)

$$D = \epsilon_0 \epsilon_r \mathcal{E} \quad \text{units of C/m}^2$$



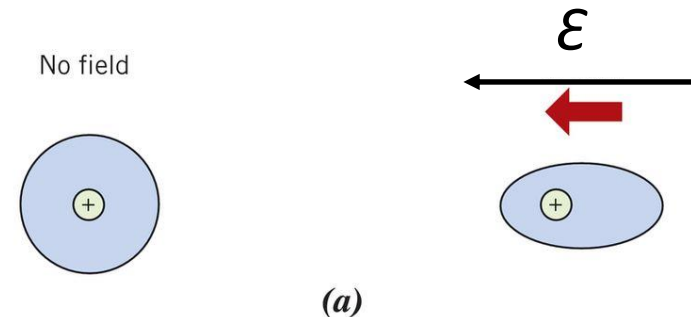
The charge density of a capacitor is much greater when a polarizable dielectric material is used as the insulator

Schematic representations of (a) the charge stored on capacitor plates for a vacuum, and (b) the increased charge-storing capacity resulting from the polarization of a dielectric material.

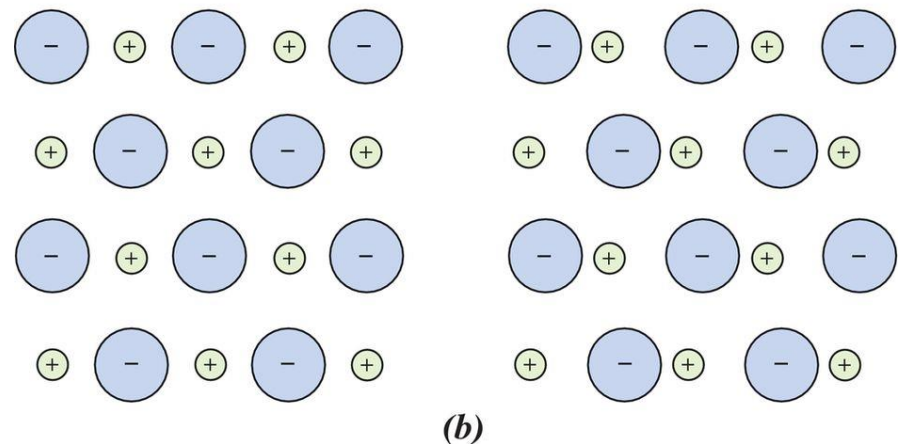
# Types of polarization

- 3 types of polarization

**1) Electrical polarization,**  
distortion of the  
electron cloud



**2) Ionic polarization,**  
displacement of the  
ions in the crystal



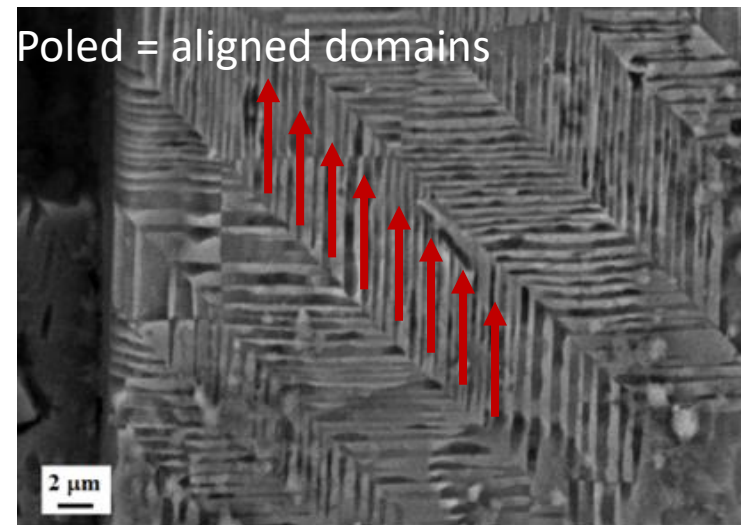
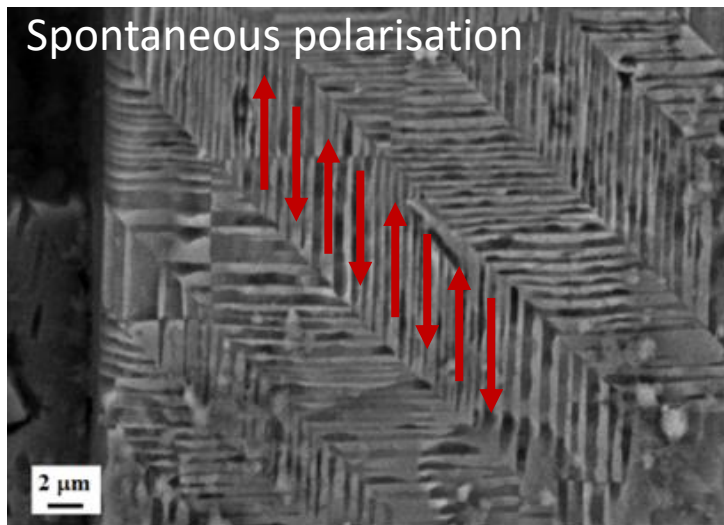
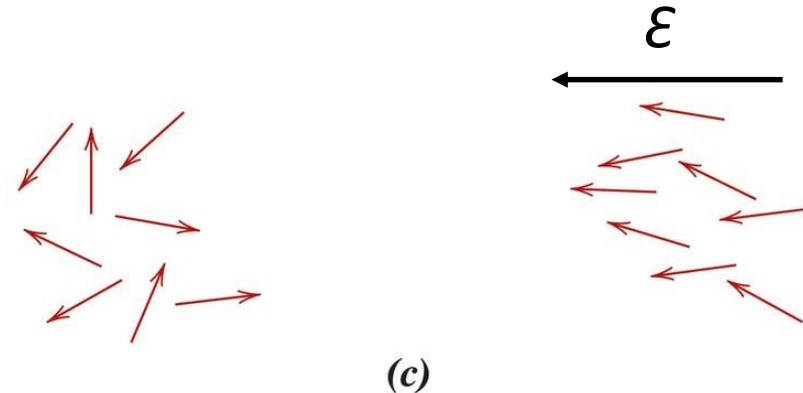
(a) Electronic polarization that results from the distortion of an atomic electron cloud by an electric field.  
(b) Ionic polarization that results from the relative displacements of electrically charged ions in response to an electric field. Fig 19.32 Callister & Rethwisch 9e

# Types of polarization

- 3 types of polarization

## 3) Orientation polarization, alignment of dipole domains with the field

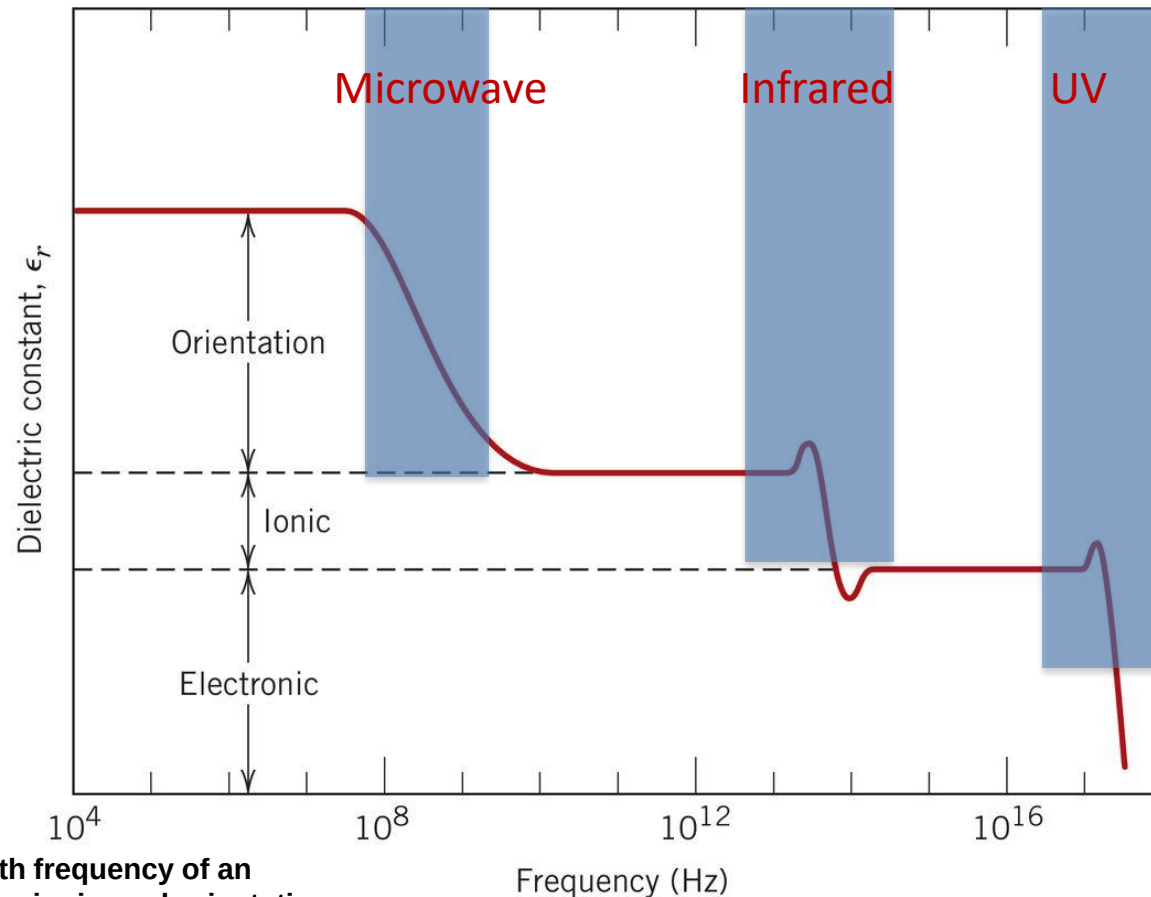
Example  $\text{BaTiO}_3$



High dielectric constant,  $k$ , 2000-10,000 at 1Mhz

# Frequency dependence of $\epsilon_r$

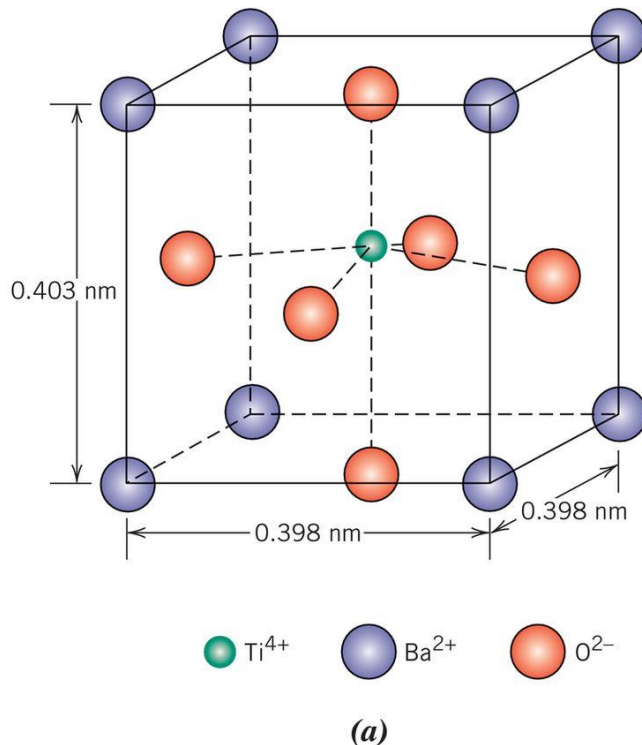
- frequency dependence due to speed at which dipoles can switch
- Fastest for electronic polarization, slowest for orientation polarization



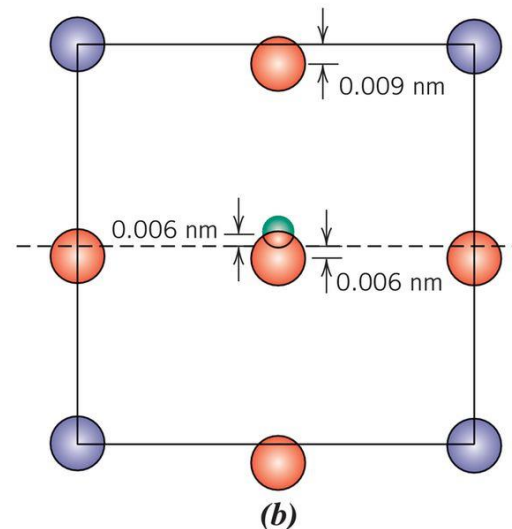
Variation of dielectric constant with frequency of an alternating electric field. Electronic, ionic, and orientation polarization contributions to the dielectric constant are indicated. Fig 19.34 Callister & Rethwisch 9e

# Ferroelectric behaviour

- **ferroelectric** materials are those that spontaneously polarize [ionic polarization] below their Curie Temperature,  $T_c$   
For  $\text{BaTiO}_3$  ( $T_c \sim 120^\circ\text{C}$ )



Displacements are small (pm)  
but polarization of charge is large



A barium titanate ( $\text{BaTiO}_3$ ) unit cell (a) in an isometric projection, and (b) looking at one face, which shows the displacements of  $\text{Ti}^{4+}$  and  $\text{O}^{2-}$  ions from the center of the face. Fig 19.35 Callister & Rethwisch 9e

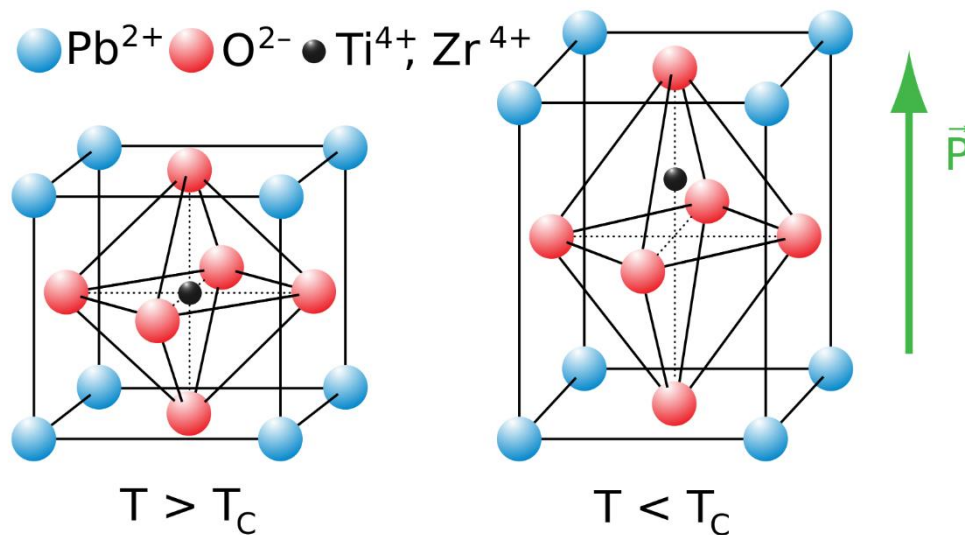
# Piezoelectric properties

Piezoelectric materials generate a **voltage** in response to a **mechanical strain**

The distortion of the crystal lattice results in ionic polarization

**All ferroelectric materials are piezoelectric**

But **not all piezoelectric materials are ferroelectric**



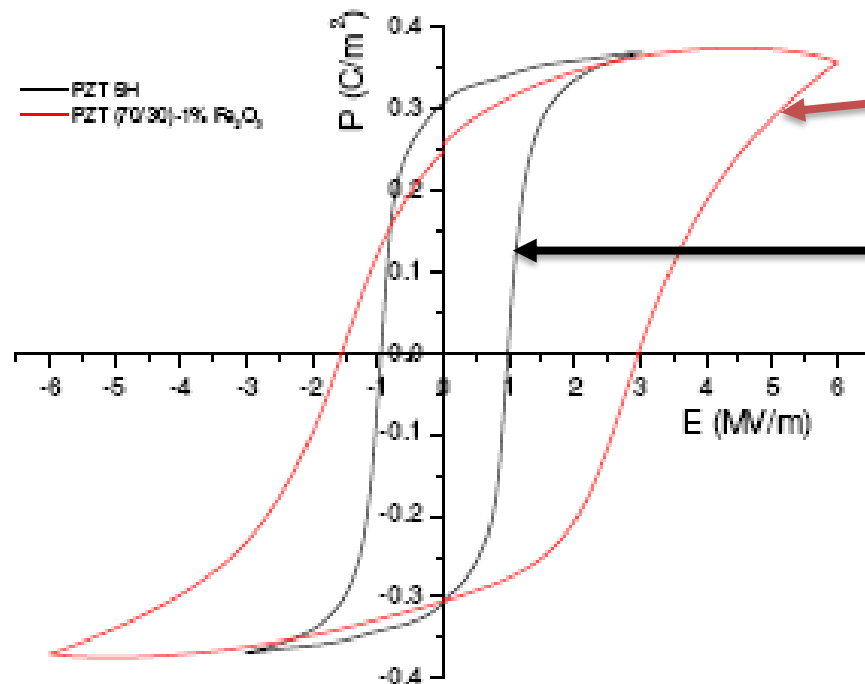
Lead zirconate titanate (PZT)  
 $\text{Pb}[\text{Zr}_x \text{Ti}_{1-x}]\text{O}_3$

Perovskite crystal structure

# Piezoelectric materials

Piezoelectric materials **polarize** in response to an **electric field**

The P-E response is **hysteretic** because of the switching of the ferroelectric domains (orientation polarization)



## Hard ferroelectric:

- difficult switching
- memory applications

## Soft ferroelectric:

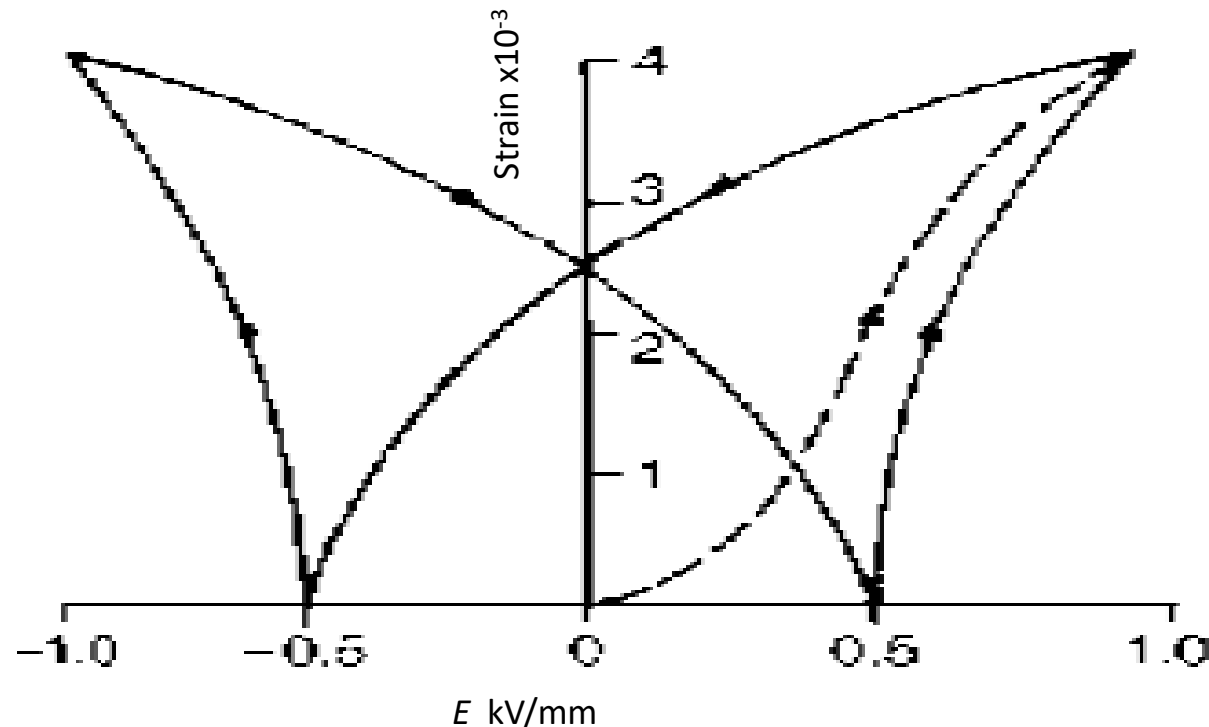
- easy switching
- resonator applications

Polarization – electric field response for PZT

**P-E loop**

# Piezoelectric materials

The converse piezoelectric effect – they generate **strain** in response to an **applied voltage**. The behaviour is also hysteretic and so requires a reverse field to restore the material to zero strain



Strain – electric field response for PZT  
**S-E loop or Butterfly loop**



## Dielectric behaviour:

Ability to transmit an electric force without conduction

Higher dielectric constant than air (or vacuum) improves the charge density of capacitors

Materials can polarize in 3 ways

Ferroelectric materials spontaneously polarize below  $T_c$

Piezoelectric materials polarize in response to a strain